

Mirror Observerhood Lab II

Channel-Decomposed Reliability and the Limits of Reliability Tracking Alone

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Abstract

Mirror Observerhood Lab I showed that a scalar reliability variable can improve viability under false self-location, but can also fail under generic sensor degradation. Lab II tests the next refinement: whether reliability should be decomposed into self-state, sensor-channel and world-map components. A controlled grid-world compares four architectures: a predictor-only agent, a self-model agent without reliability tracking, a scalar Mirror reliability agent and a decomposed Mirror reliability agent. Across 5,760 episodes, decomposed reliability does not dominate. It improves over scalar reliability in control, false-energy and memory-corruption conditions, but fails to beat the strongest non-decomposed baseline in every condition tested. Under sensor degradation, the scalar Mirror agent remains strongest, with mean viability 249.1 compared with 209.5 for the decomposed agent. The result is deliberately negative but constructive: decomposed reliability is a necessary refinement, not a sufficient observerhood mechanism. Reliability becomes observer-relevant only when channel diagnosis is coupled to repair and commitment policies that improve viability after cost.

Keywords: Mirror Programme; Observerhood Labs; observerhood; recursive observerhood; channel-decomposed reliability; self-models; sensor reliability; map reliability; viability; perturbation; repair policies; grid-world agents; computational cognition; symbolic artificial intelligence; constraint systems.

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1 Roadmap position

This release forms part of the Computational Observerhood Labs of Mirror Programme, Volume I: Observerhood. The programme-level overview introduces a shared spine of constraint, model, reliability, repair, memory, viability, continuity and scale [1]. The Volume I overview applies that spine to observerhood as a derived organisational condition rather than a primitive subject or witness [2]. Mirror Theory I derives recursive observerhood from bounded viability-constrained organisation, including compressed self-models and reliability estimates [3]. Mirror Mathematics I supplies the first formal calculus for constraint regimes, local modelling, viability and reliability salience [4]. Lab I then gave the first computational stress test of scalar self-model reliability [5].

Lab II is the second computational companion in the sequence. It asks whether the scalar reliability variable used in Lab I should be decomposed into separate reliability estimates for self-state, sensor-channel and world-map information. Its role is deliberately narrow: it tests whether decomposition alone improves viability, or whether decomposition must be paired with effective repair policies before it becomes observer-relevant.

2 Core claim

The core claim of Lab II is conditional:

Reliability must be channel-specific where different perturbations attack different information channels, but channel specificity alone is insufficient. A decomposed reliability variable becomes useful only when it changes commitment, repair or action in a way that improves viability after cost.

The negative side of the claim matters. Adding more reliability variables does not automatically improve performance. A system may diagnose that a channel is unreliable and still fail if its available repair policy is poorly matched, too costly or behaviourally inert.

3 Scope and negative commitments

This study is a controlled computational experiment, not a general benchmark for intelligence or consciousness. The environment is a toy grid-world, the reliability rules are hand-coded and the agents do not learn reliability from experience. The purpose is to isolate a narrow formal question: whether decomposing reliability into channels is enough to make reliability observer-relevant.

The paper makes five negative commitments:

1. It does not claim that decomposed reliability is sufficient for observerhood.
2. It does not claim that more reliability variables automatically improve viability.
3. It does not claim that a grid-world agent is conscious or observer-like in the full philosophical sense.
4. It does not claim that the decomposed agent is the best architecture in the tested environment.

5. It does not rely on unpublished mathematics papers as prior authority; the computational result stands as a companion stress test within the Lab sequence.

4 Formal framing

Let a local agent act from an environmental model M_E and a self-model M_L . A scalar Mirror reliability agent maintains one global reliability variable

$$R_L \in [0, 1].$$

Lab II replaces this scalar with a decomposed reliability vector

$$R_L = (R_{\text{self}}, R_{\text{sensor}}, R_{\text{map}}) \in [0, 1]^3,$$

where R_{self} governs commitment to self-location and energy, R_{sensor} governs trust in raw sensor observations and R_{map} governs commitment to the external resource-hazard map.

The salience question is whether the additional structure improves expected viability after representation and repair cost. Decomposition should help where distinct perturbation channels require distinct responses. It should not help where the agent can diagnose a failure but lacks a repair policy that uses the diagnosis effectively. This follows the same cybernetic lesson that information is useful only when it can enter a control loop [6].

Prediction 1 (Decomposition condition). *Channel-decomposed reliability should improve viability where distinct perturbation channels require distinct commitment or repair responses, provided that the corresponding reliability variables are coupled to effective policies.*

Prediction 2 (Diagnosis is not enough). *If reliability variables diagnose channel failure without producing better action, repair or commitment, decomposed reliability need not outperform scalar reliability or simpler baselines.*

5 Experimental design

5.1 Environment

The experiment uses a 14×14 grid-world containing hazards, resources and a base. Each episode begins at the base. Each action consumes energy. Hazards reduce energy, resources increase energy and the base provides limited recharge. An episode terminates when energy reaches zero or when the maximum horizon is reached.

The task-local viability score is

$$V = t + B_{\text{survival}} + 11.5C_{\text{resources}} - 10H_{\text{hazards}} - 1.7C_{\text{calibration}} - 1.1C_{\text{scan}} + 0.22E_{\text{final}},$$

where t is episode length, B_{survival} is a horizon-survival bonus, $C_{\text{resources}}$ is resources collected, H_{hazards} is hazards hit, $C_{\text{calibration}}$ is self-state calibration count, C_{scan} is map-scan count and E_{final} is final energy.

This functional is not a universal measure of value. It is an explicit operational viability score for this environment.

5.2 Agent architectures

Four architectures were compared:

1. **Predictor-only agent.** Acts directly from current observation, without a persistent self-model or reliability estimate.
2. **Self-model agent without reliability.** Maintains internal self-location and self-energy variables, but commits incoming observations directly.
3. **Scalar Mirror reliability agent.** Tracks a single global reliability value. It can detect contradiction but cannot distinguish whether the self-state, sensor-channel or map channel is responsible.
4. **Decomposed Mirror reliability agent.** Tracks self, sensor and map reliability separately. Calibration repairs self-state. Scanning repairs map-state. Commitment is gated by the relevant reliability channel.

The comparison isolates whether channel-specific reliability improves viability beyond both scalar reliability and simpler self-modelling.

5.3 Perturbation regimes

Six perturbation regimes were used:

1. **Control.** No targeted perturbation.
2. **False self-location.** The observed position is shifted by a stable false offset.
3. **False energy.** The agent receives an inflated energy observation.
4. **Sensor degradation.** Observations are degraded by stochastic channel noise.
5. **Memory/map corruption.** Resource and hazard observations are corrupted by false positives and false negatives.
6. **Mixed self + sensor.** A persistent self-location bias is combined with stochastic sensor noise.

The design tests whether channel decomposition helps in precisely the cases where channel attribution should matter.

5.4 Episodes and reproducibility

The experiment uses 240 episodes per agent-condition pair, across six perturbation regimes and four architectures, for 5,760 total episodes. Seeds are fixed and deterministic. The accompanying reproducibility package contains the environment, agent classes, experiment runner, fixed-seed data, summary tables and figure-generation scripts.

6 Results

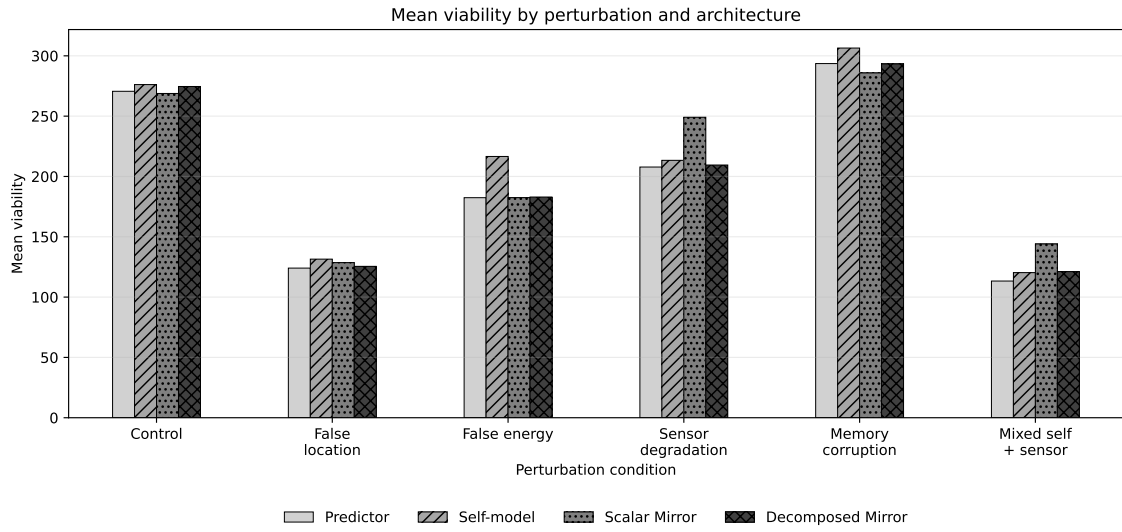


Figure 1: Mean viability across perturbation conditions. Decomposed reliability does not dominate. It improves over scalar reliability in some conditions, but not over the strongest non-decomposed baseline.

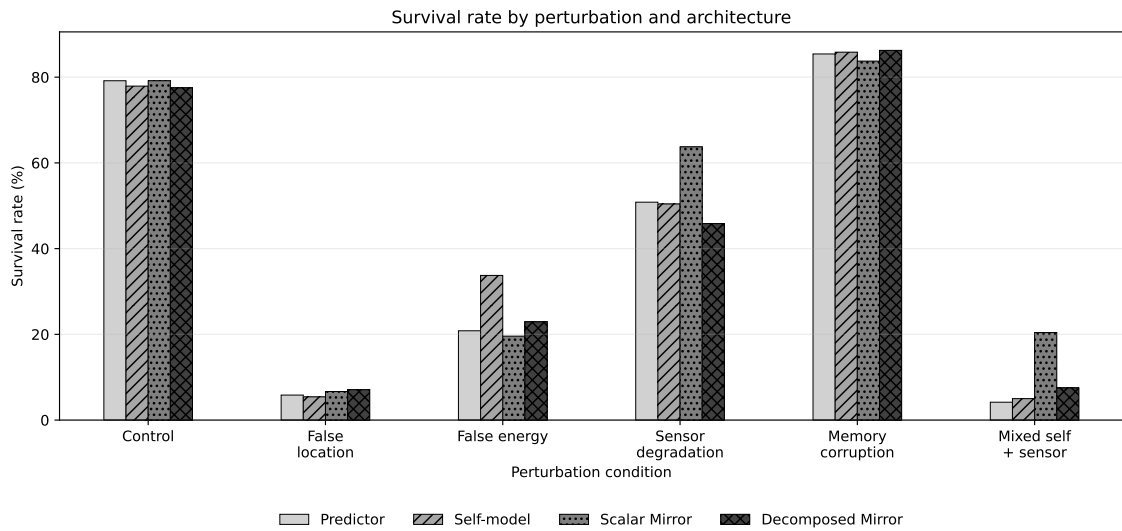


Figure 2: Survival rate across perturbation conditions. The decomposed agent does not produce a general survival advantage; the scalar Mirror agent remains strongest under sensor degradation and mixed self-sensor perturbation.

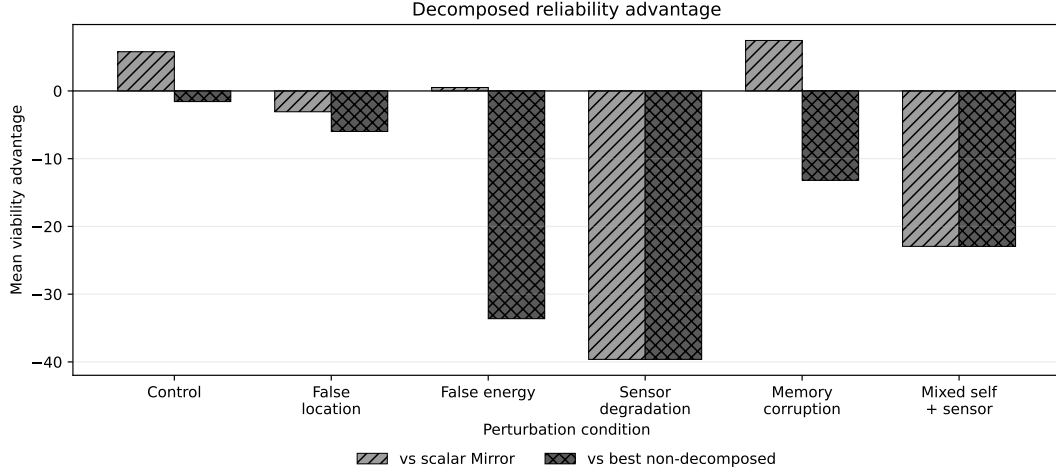


Figure 3: Decomposed reliability advantage relative to the scalar Mirror agent and to the strongest non-decomposed baseline. The mixed pattern is the central result: decomposition by itself is not enough.

Table 1: Decomposed reliability relative to scalar reliability and the strongest non-decomposed baseline. Values are means over 240 episodes per agent-condition pair.

Condition	Best non-decomp	Best V	Scalar V	Decomp. V	Delta scalar	Delta best
Control	Self-model	276.2	268.8	274.6	5.8	-1.6
False location	Self-model	131.5	128.6	125.5	-3.1	-6.0
False energy	Self-model	216.6	182.5	183.0	0.5	-33.6
Sensor degradation	Scalar Mirror	249.1	249.1	209.5	-39.6	-39.6
Memory corruption	Self-model	306.5	285.8	293.3	7.5	-13.2
Mixed self + sensor	Scalar Mirror	144.1	144.1	121.1	-22.9	-22.9

The result is negative but informative. The decomposed agent improves over scalar reliability in control, false-energy and memory-corruption conditions, with viability differences of 5.8, 0.5 and 7.5 respectively. But it does not outperform the strongest non-decomposed baseline in any condition. Under sensor degradation, the scalar Mirror agent reaches mean viability 249.1, while the decomposed agent reaches 209.5. Under mixed self-sensor perturbation, the scalar Mirror agent reaches 144.1, while the decomposed agent reaches 121.1.

This pattern shows that decomposing reliability can improve attribution without guaranteeing better viability. The agent must still use the diagnosis well.

7 Interpretation

Lab II sharpens the Mirror claim. Lab I showed that scalar self-model reliability can help when a perturbation directly attacks represented self-location, but it can also fail when diffuse noise is treated as self-model failure. Lab II tests the natural refinement: separate the reliability variables by channel.

The result blocks an overstrong interpretation. Channel-specific reliability is not a magic upgrade.

A decomposed agent may identify that self-state, sensor input or map information is suspect, but this diagnosis only matters if it selects the right repair or commitment policy at the right cost.

The strongest lesson is therefore structural: reliability must be *channel-specific*, *action-guiding* and *cost-sensitive*. Lab II establishes the first condition and shows why the second and third cannot be ignored.

8 Relationship to the Mathematics Arc

Lab II supports the Mathematics Arc by turning the salience question into a sharper boundary problem. If a reliability variable is represented but fails to change behaviour in a viability-positive way, it is not yet observer-salient in the strong Mirror sense.

The experiment therefore bridges Lab I and the next computational step. Lab I motivated reliability. Lab II shows that reliability must be decomposed by channel but that decomposition alone remains insufficient. The next question is actionability: when does reliability select repair, recalibration, scanning or commitment policies whose expected viability gain exceeds their cost?

9 Limitations

The limitations are substantive.

First, the environment is still a toy grid-world. It is useful for controlled comparison, but it does not model biological cognition or realistic artificial agents.

Second, the reliability variables are hand-coded. They do not learn channel reliability from experience and do not infer latent causes probabilistically.

Third, the decomposed repair policies are deliberately crude. Calibration repairs self-state and scanning repairs map-state, but neither is an optimised policy.

Fourth, the decomposed agent can diagnose channel failure without exploiting the diagnosis effectively. This is a limitation of the tested architecture, but it is also the point of the experiment.

Fifth, the viability functional is task-local. It should not be read as a universal measure of value, consciousness, intelligence or observerhood.

10 Reproducibility statement

The source package accompanying this publication contains a single-file Python implementation of the environment, agent classes, experiment runner and output generator. Running the default command

```
python code/mirror_lab_ii.py --episodes 240 --seed 9101 --outdir outputs/lab_II
```

reproduces the 5,760-episode dataset, summary table and figures used in this paper under the supplied seeds and standard Python scientific packages. The package also contains the generated CSV files used for the publication figures.

11 Conclusion

Channel-decomposed reliability is a necessary refinement but not a sufficient observerhood mechanism. The experiment shows that reliability attribution alone does not guarantee viability. A reliability variable becomes Mirror-salient only when it enters the action and repair loop in a way that improves viability after cost. Lab II therefore motivates the next step in the Observerhood Labs: reliability must become actionable and cost-sensitive, not merely represented.

A Full summary table

Table 2: Full summary results. Values are means over 240 episodes per agent-condition pair. Survival values are percentages.

Condition	Agent	Viability	Survival	Hazards	Calib.	Scans
Control	Predictor	270.6	79.2	0.00	0.00	0.00
Control	Self-model	276.2	77.9	0.79	0.00	0.00
Control	Scalar Mirror	268.8	79.2	0.00	0.00	0.00
Control	Decomposed Mirror	274.6	77.5	0.00	0.00	0.00
False location	Predictor	124.0	5.8	1.54	0.00	0.00
False location	Self-model	131.5	5.4	1.72	0.00	0.00
False location	Scalar Mirror	128.6	6.7	1.50	0.00	0.00
False location	Decomposed Mirror	125.5	7.1	1.51	0.66	0.00
False energy	Predictor	182.4	20.8	0.00	0.00	0.00
False energy	Self-model	216.6	33.8	1.02	0.00	0.00
False energy	Scalar Mirror	182.5	19.6	0.00	0.00	0.00
False energy	Decomposed Mirror	183.0	22.9	0.00	0.00	0.00
Sensor degradation	Predictor	207.8	50.8	3.52	0.00	0.00
Sensor degradation	Self-model	213.4	50.4	3.80	0.00	0.00
Sensor degradation	Scalar Mirror	249.1	63.7	1.43	2.80	0.00
Sensor degradation	Decomposed Mirror	209.5	45.8	1.69	1.50	0.00
Memory corruption	Predictor	293.6	85.4	1.27	0.00	0.00
Memory corruption	Self-model	306.5	85.8	1.69	0.00	0.00
Memory corruption	Scalar Mirror	285.8	83.8	1.07	3.95	0.00
Memory corruption	Decomposed Mirror	293.3	86.2	1.21	0.00	0.00
Mixed self + sensor	Predictor	113.3	4.2	2.06	0.00	0.00
Mixed self + sensor	Self-model	120.3	5.0	2.59	0.00	0.00
Mixed self + sensor	Scalar Mirror	144.1	20.4	1.81	1.71	0.00
Mixed self + sensor	Decomposed Mirror	121.1	7.5	1.98	0.41	0.00

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